Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC *Hack*athon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

Prompt Masters

Project / Solution Name

EntreMaster

Name (Leader)

Lian Yi Yen

Email & Phone (Leader)

yiyenlian7@gmail.com & 011-5620 9436

University Affiliation(s)

University Malaya

Live Prototype URL

Code Repository URL (public / access granted to judges)

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

EntreMaster adopts a secure 3-tier architecture. The frontend is built as a single-page app (HTML5/JS) communicating with a Node.js/Express REST backend. The backend handles request routing, authorization, and session verification via custom security layers. Persistent inventory data is stored in PostgreSQL (Supabase / local PG17) utilizing Row-Level Security (RLS) for tenant isolation, coupled with connection pooling (pg/node-postgres) and SSL enforcement (CA certificates). A Python micro-service handles offline security automated testing. End-to-end user identity is validated against explicit JWT/role boundaries before database transactions occur, guaranteeing multi-tenant security and zero unauthorized data access.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Node.js (v18+) & Express (v4.x)	Core API Server & Request Routing	MIT
JavaScript (ES6+) / HTML5 / CSS3	Web Frontend User Interface	Open Source / Native
pg / node-postgres (v8.x)	PostgreSQL Database Client & Connection Pooling	MIT
jsonwebtoken (v9.x)	Authentication & Tenant Token Verification	MIT
Python (v3.10+) & pytest	Security Audit & Offline Test Suite Execution	PSL/MIT
dotenv (v16.x)	Environment Variable & Credentials Management	BSD-2-Clause

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Supabase Database API	Managed PostgreSQL host & Auth API endpoints	Free Tier

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Web Server Hosting, Node.js runtime & Replit Secret store	Free / Standard
Supabase Platform	Cloud Managed PostgreSQL Database & SSL Endpoint	Free Tier

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
None	-	-	-

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
EntreMaster Test Inventory Dataset	Internal / Self-generated	Synthetic	MIT / Open	N

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Replit Agent	Boilerplate code drafting, architecture refactoring, and test script generation	Major
ChatGPT	Security review planning, RLS policy verification, and documentation writing	Moderate

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

All AI-assisted code, security policies, and architectural designs were manually audited, refactored, and verified by the human team. We implemented an offline test runner (run-phase2-isolated.py & security.test.cjs) to conduct strict automated unit and integration tests against local PostgreSQL instances, validating every RLS policy, tenant boundary, and transaction fallback to ensure full security compliance and functional correctness.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☑ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
N/A	-	-

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

No official sponsor-provided resources were used for this project; the solution runs entirely on standard open-source tools and public free-tier infrastructure.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

EntreMaster collects minimal operational user data (username, role, tenant ID) solely for authentication and multi-tenant inventory management. Passwords/tokens are never stored in plain text. All database transport is encrypted using SSL/TLS with Supabase CA certificates. PostgreSQL Row-Level Security (RLS) strictly isolates tenant data, ensuring no user can read or modify another tenant's inventory records. Session tokens (JWTs) are short-lived. No personal data is shared with third-party tracking or external GenAI APIs.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Free-Tier Cold Starts / Database Sleep: Supabase free-tier databases may pause after inactivity, causing initial latency. Mitigation: Connection pooling and retry mechanisms are implemented in backend pools.

Environment Dependency: Relies on Replit secrets for DATABASE_URL and JWT_SECRET. Mitigation: Fallback error handlers ensure secure failure without leaking credentials or internal stack traces.

Rate Limits: Cloud hosts place connection rate limits. Mitigation: Restricted pool sizing (max: 10-20) in Node.js client configurations.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

☑ Any use of official SDC sponsor resources has been declared in Section 5.

☑ The prototype described here is accessible to judges through the links provided in Section 1.

☑ All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature] 

[Date] 25/10/2026